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SECJ 3303 – INTERNET PROGRAMMING

TOPIC 5 – JAVA WEB SESSION TRACKING

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OBJECTIVES

Applied

- Provide for session tracking by using cookies, HTTP session and URL encoding.
- Provide for parameter passing by using URL rewriting and hidden fields.
- Test your web applications with cookies enabled and with cookies disabled.
- Write a utility class that includes a static method for getting a specific cookie from a user's browser.

Knowledge

- Describe the way HTTP works without session tracking.
- Describe the way cookies and HTTP session are used for session tracking.
- Describe the way URL encoding is used for session tracking.
- Distinguish between persistent cookies and per-session cookies.
- Distinguish between the use of URL rewriting and the use of hidden fields as ways to implement parameter passing.
- Describe the way HTTP session in servlet and JSP are used for session tracking.

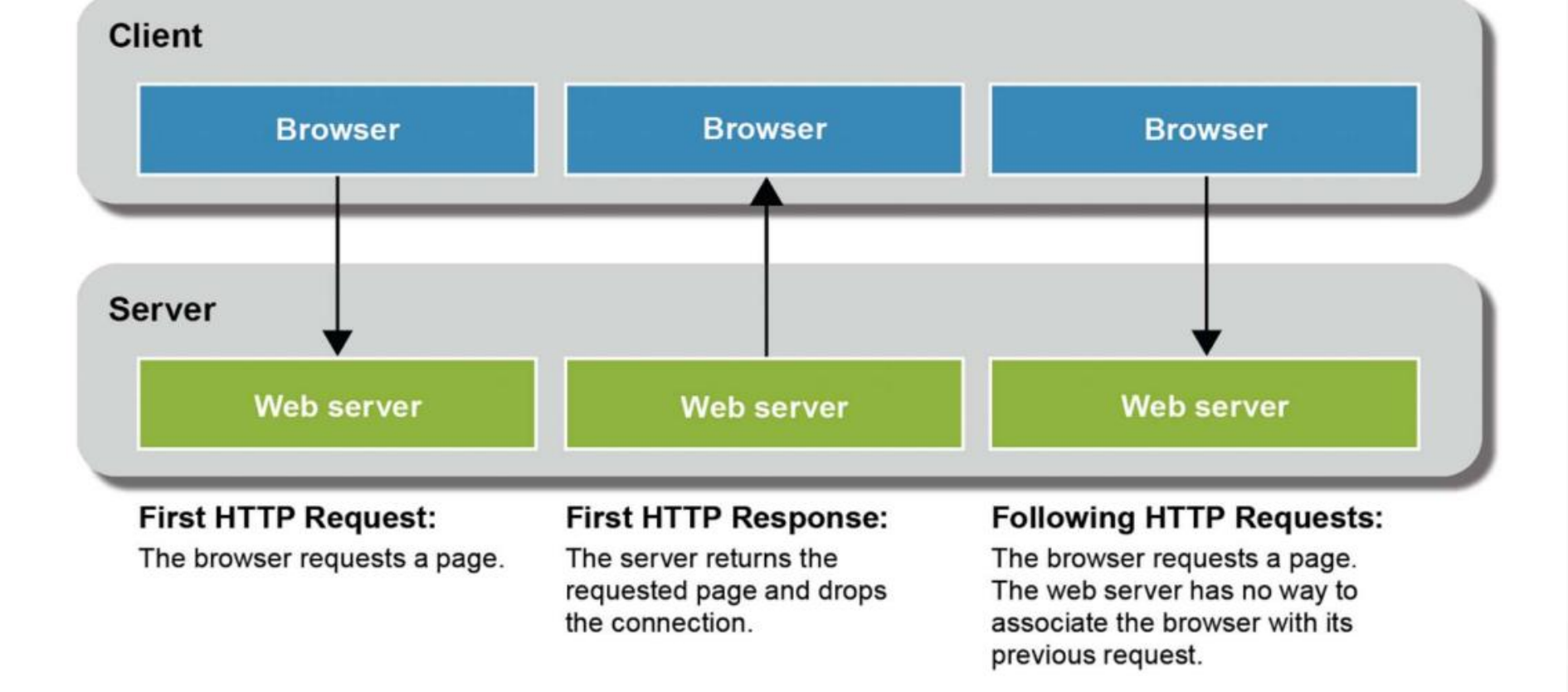
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Session Tracking

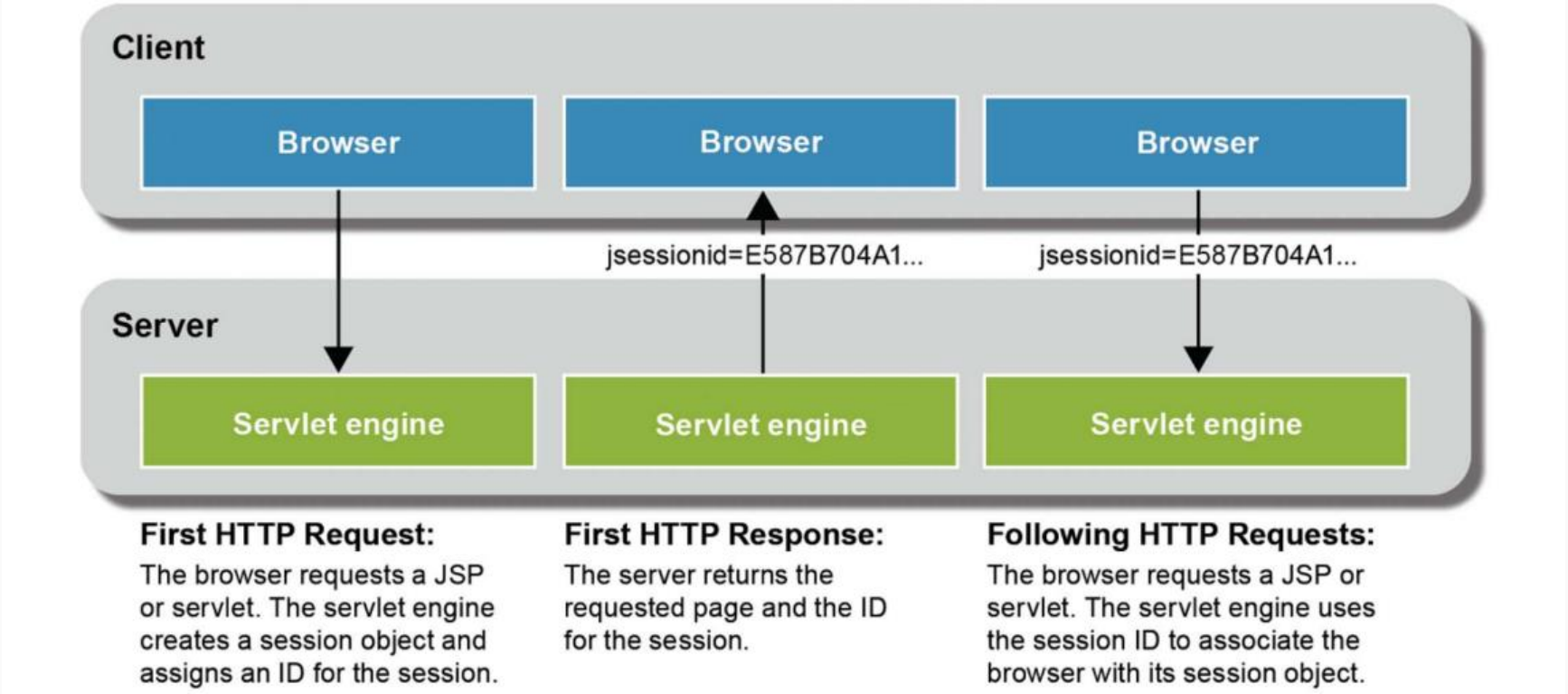
- **Session** is a particular interval of time.
- **Session Tracking** is a mechanism used by the **Web container** to store session information for a particular user.
- It is also known as a **Session Management** which is a way to maintain state (data) of the user.
- HTTP is a ***stateless protocol*** which means each request is considered as the new request. Once a browser makes a request, it drops the connection to the server. So, to **maintain *state***, a web application must use *session tracking*.

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Why session tracking is difficult with HTTP?



How Java keep tracks of session?

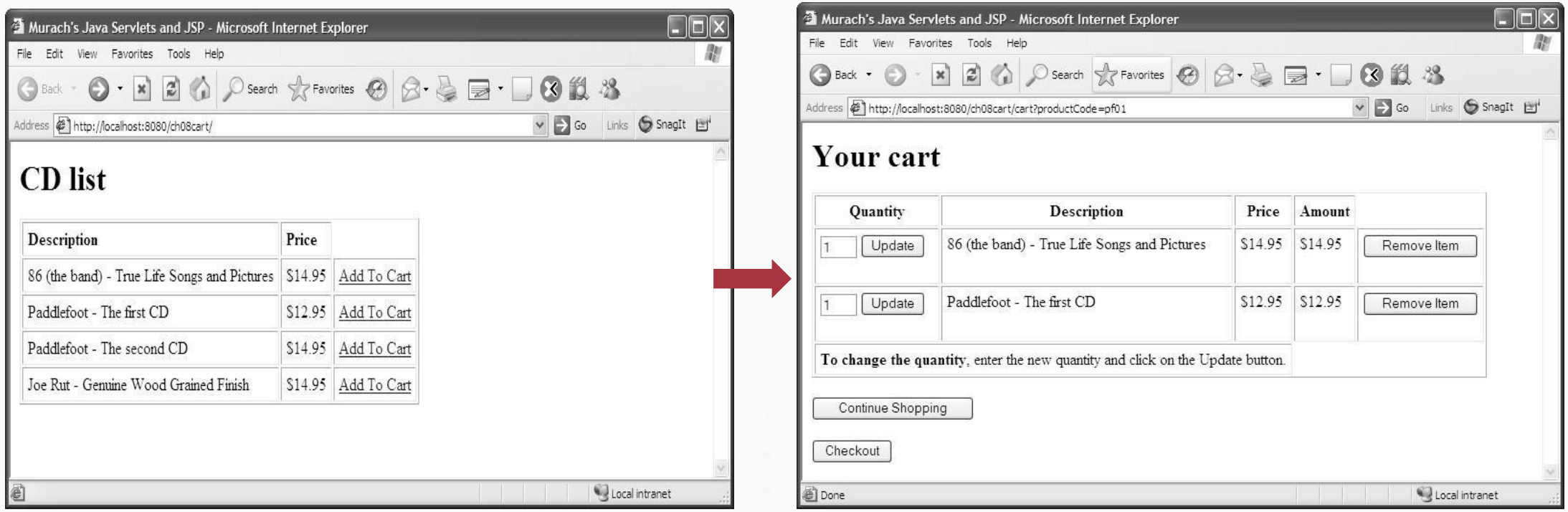


How Java keep tracks of session?

- A browser on a client request a JSP or servlet from the web server, which passes the request to the servlet engine.
- Then, the servlet engine checks if the request includes an ID for the Java session.
- If it doesn't, the servlet engine **creates a unique ID for the session plus a session objec**t that can be used to store the data for the session.
- From that point on, the web server uses the session ID to relate each browser request to the session object, even though the server still drops the HTTP connection after returning each page.

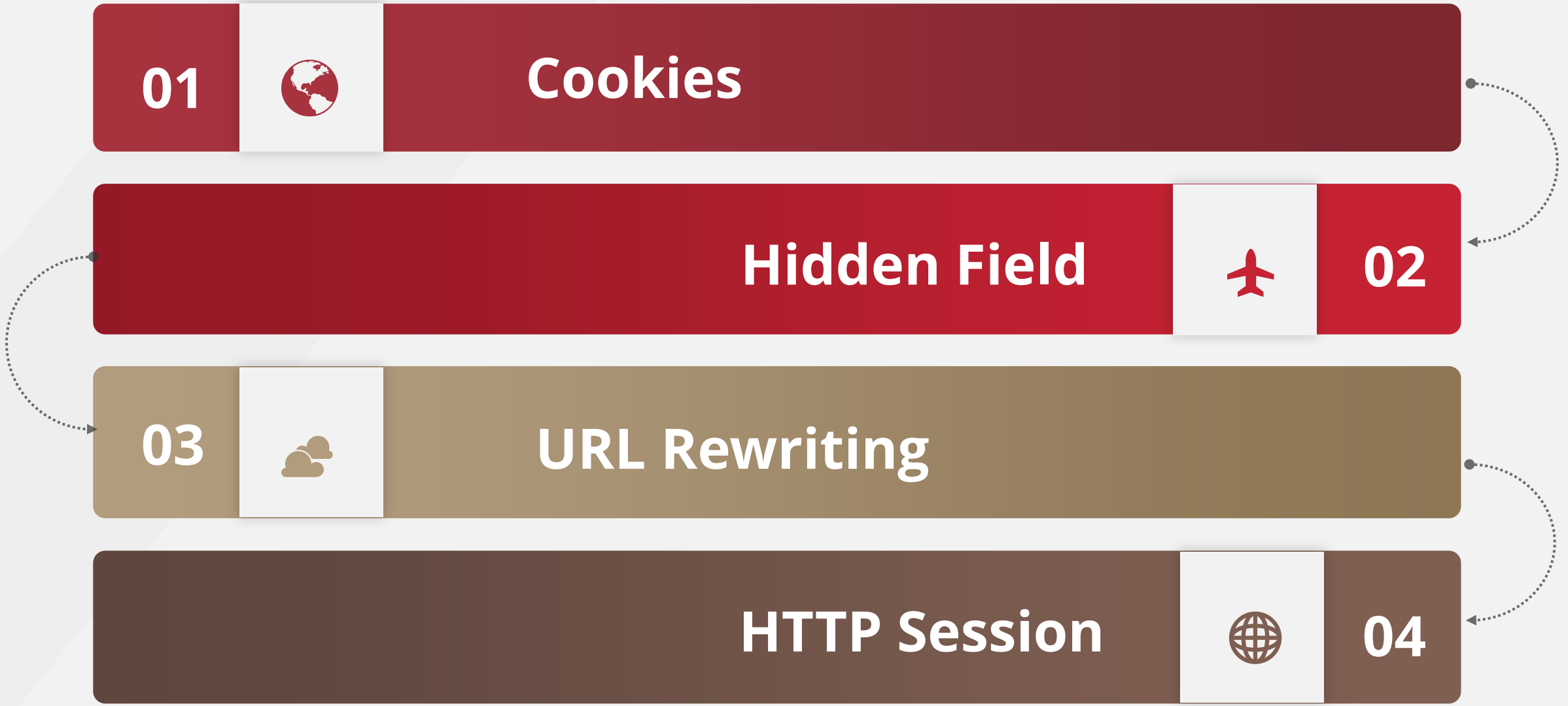
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Example of session tracking



For example, a shopping cart module should know: who is sending the request to add an item, in which cart the item has to be added or who is sending checkout request. So that, it can charge the amount to the correct client.

Techniques for Session Tracking



Cookies

- A **cookie** is a small piece of information that is persisted between the multiple client requests.
- A cookie has a name, a single value, and optional attributes such as a comment, path and domain qualifiers, a maximum age, and a version number.
- By default, the servlet API uses a *cookie* to store a session ID in each browser. Then, the browser passes the cookie to the server with each request. To store the data for each session, the server creates a *session object*.
- There are two types of cookies:
  - *persistent cookies* are store on the user's PC. It is valid for multiple session . It is not removed each time when user closes the browser. It is removed only if user logout or sign-out.
  - *per-session cookies (non-persistent)* are deleted when the session ends. It is valid for single session only. It is removed each time when user closes the browser.

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How cookies work?

- A cookie is a name/value pair that is stored in a browser.
- On the server, a web application creates a cookie and sends it to the browser.
- On the client, the browser saves the cookie and sends it back to the server every time it accesses a page from that server.
- Cookies can be set to persist within the user's browser for up to 3 years.
- Some users disable cookies in their browsers. As a result, you can't always count on all users having their cookies enabled.
- Browsers generally accept only 20 cookies from each site and 300 cookies total. In addition, they can limit each cookie to 4 kilobytes.
- A cookie can be associated with one or more subdomain names.

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Typical uses for cookies

- **To allow users to skip login and registration forms** that gather data like user name, password, address, or credit card data.
- **To customize pages** that display information like weather reports, sports scores, and stock quotations.
- **To focus advertising** like banner ads that target the user's interests.

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Cookies

Constructor of the Cookie class

Constructor	Description
`Cookie(String name, String value)`	Creates a cookie with the specified name and value.

A method of the response object

Method	Description
`addCookie(Cookie c)`	Adds the specified cookie to the response.

A method of the request object

`getCookies()`	Returns an array of Cookie objects that the client sent with this request. If no cookies were sent, this method returns a null value.

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The methods of the Cookie class

Cookies

Method	Description
setMaxAge(int maxAgeInSeconds)	To create a persistent cookie, set the cookie's maximum age to a positive number. To create a per- session cookie, set the cookie's maximum age to –1. Then, the cookie will be deleted when the user exits the browser.
setPath(String path)	To allow the entire application to access the cookie, set the cookie's path to "/".
getName()	Returns a string for the name of the cookie.
getValue()	Returns a string that contains the value of the cookie.

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Code that creates and sets a cookie

```
Cookie userIdCookie = new Cookie("userIdCookie", userId);
userIdCookie.setMaxAge(60*60*24*365*2);   //set the age to 2 years
userIdCookie.setPath("/");          // allow access by the entire application
response.addCookie(userIdCookie);
```

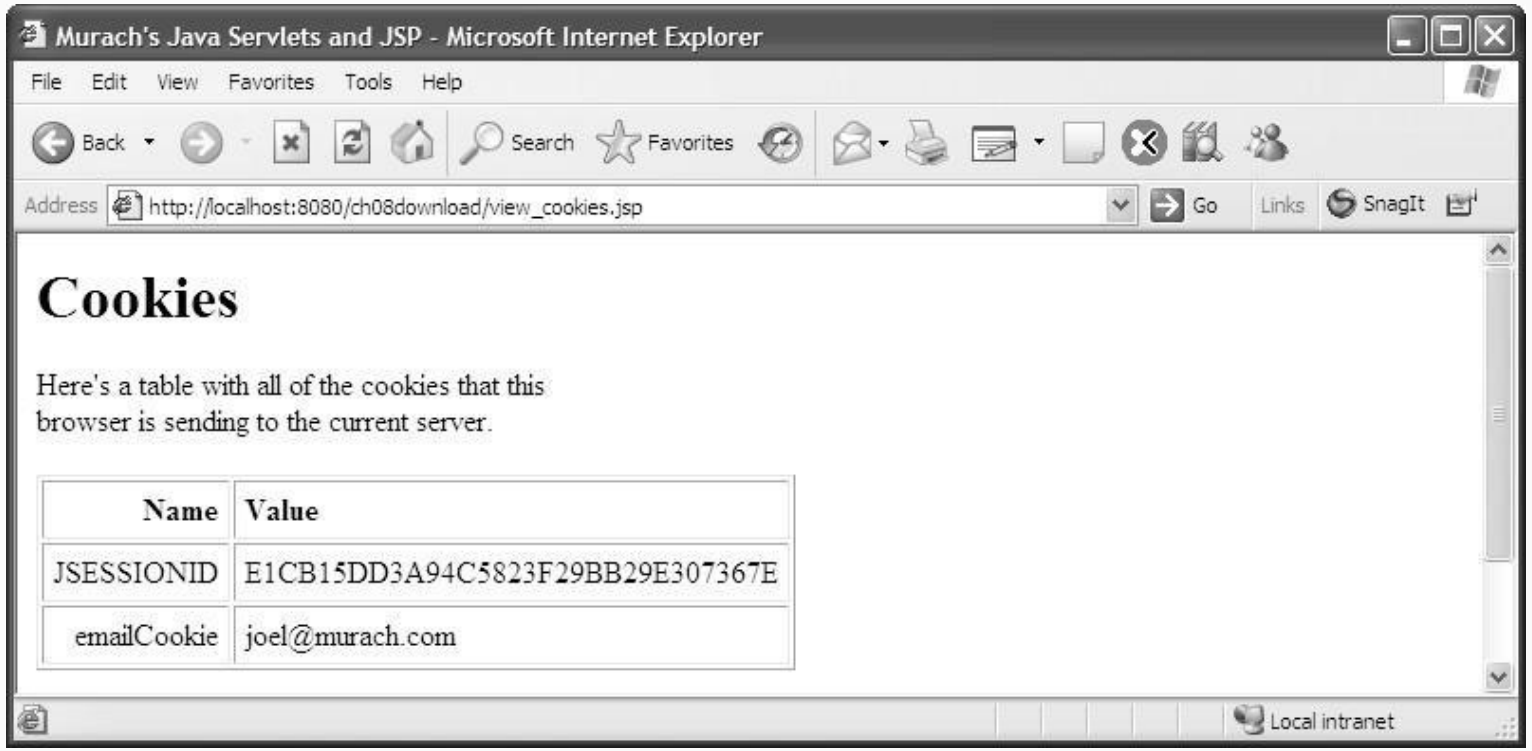
Code that gets the cookie

```
Cookie[] cookies = request.getCookies();
String cookieName = "userIdCookie";
String cookieValue = "";

for (int i=0; i<cookies.length; i++)
{
    Cookie cookie = cookies[i];
    if (cookieName.equals(cookie.getName()))
      cookieValue = cookie.getValue();
}
```


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A JSP that shows all cookies for the current server



JSP code that displays all cookies

```
<%
  Cookie[] cookies = request.getCookies();
  for (Cookie c : cookies)
  {
%>
  <tr>
    <td align="right"><%= c.getName() %></td>
    <td><%= c.getValue() %></td>
  </tr>
<%
  }
%>
```


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Servlet code that deletes all persistent cookies


```
Cookie[] cookies = request.getCookies();
for (int i=0; i<cookies.length; i++)
{
    Cookie cookie = cookies[i];
    cookie.setMaxAge(0); //delete the cookie
    cookie.setPath("/");
            //allow the entire application to access it
    response.addCookie(cookie);
}
```


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Four methods of the Cookie class

- **`setPath(`**`String path`**`)`**
- **`setDomain(`**`String domainPattern`**`)`**
- **`setSecure(`**`boolean flag`**`)`**
- **`setVersion(`**`int version`**`)`**

Note

- All of these set methods have corresponding get methods.

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A utility class that gets the value of a cookie


```
package util;import
javax.servlet.http.*;

public class CookieUtil
{
    public static String
        getCookieValue(Cookie[] cookies,
        String cookieName)
    {
        String cookieValue = "";
        Cookie cookie;
        if (cookies != null)
        {
            for (int i=0; i<cookies.length; i++)
            {
                cookie = cookies[i];
                if (cookieName.equals(cookie.getName()))
                {
                    cookieValue = cookie.getValue();
                }
            }
        }
        return cookieValue;
    }
}
```

Code that uses the CookieUtil class to get the value of a cookie

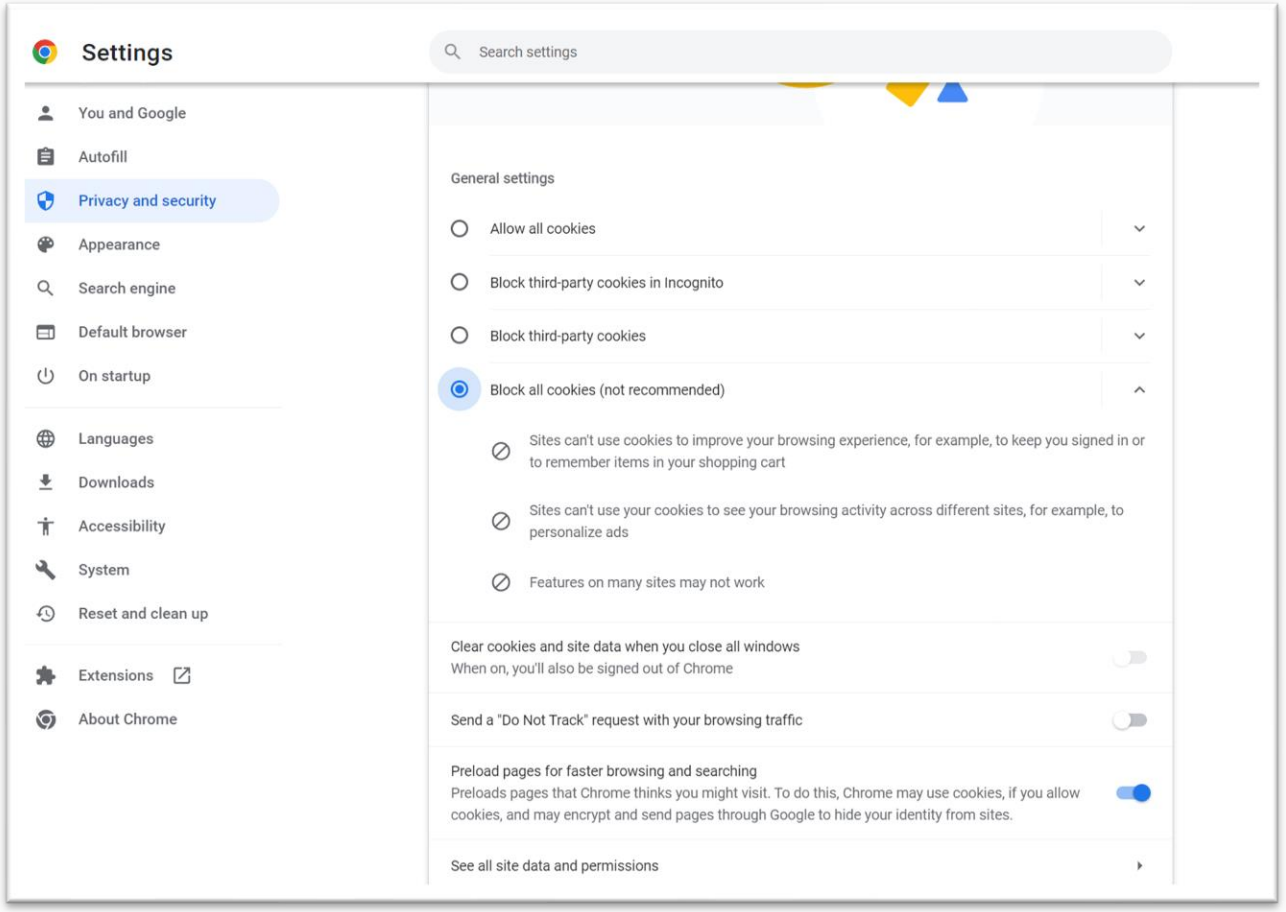
```
Cookie[] cookies = request.getCookies();
String emailAddress =
    CookieUtil.getCookieValue(cookies, "emailCookie");
```

To make it easier to get the value of a cookie, you can create a **utility class that contains a method that accepts an **array of Cookie object** and the **name of the cookie**, and then **returns the value of the cookie**.

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How to disable cookies in Google Chrome browser?

Cookies



1. On your computer, open Chrome.
2. At the top right, click More Settings.
3. Under "Privacy and security," click Site settings.
4. Then select - Cookies and other site data.
5. Select Block all cookies.

Your browser will no longer store cookies.

More info on how to disable cookies in other web browsers:
https://www.avg.com/en/signal/disable-cookies

Advantages and Disadvantages Cookies

Advantages of using cookies

- Cookies are simple to use and implement.
- Occupies less memory, do not require any server resources and are stored on the user's computer.
- We can configure cookies to expire when the browser session ends (session cookies) or they can exist for a specified length of time on the client's computer (persistent cookies).
- Cookies persist a much longer period of time than session state.

Disadvantages of using cookies

- User has the option of disabling cookies on his computer from browser's setting.
- Cookies will not work if the security level is set to high in the browser.
- Users can delete a cookies.
- User's browser can refuse cookies, so your code has to anticipate that possibility.
- Complex type of data is not allowed (e.g. dataset etc). It allows only plain text (cookie allows only string content).

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Hidden Form Field

Hidden Field

- A hidden field is used to store client state. In this case, user information is stored in hidden field value and retrieved from another servlet.
- The server embeds new hidden fields in every dynamically generated form page for the client.
- When the client submits the form to the server, the hidden fields identify the client.
- Web server can send a hidden HTML form field along with a unique session ID as follows:

```
<input type = "hidden" name = "sessionid" value = "12345">
```

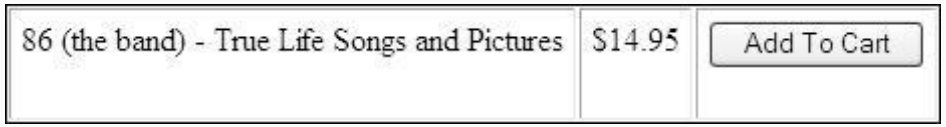

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Example of Hidden Form Field

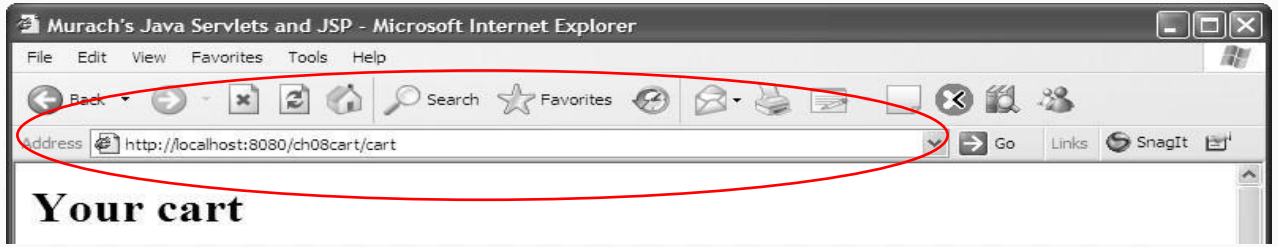
A Form tag that uses a hidden text field

```
<form action="cart" method="post">
  <input type="submit" value="Add To Cart">
  <input type="hidden" name="productCode" value="8601">
</form>
```

The form displayed in a browser



The URL that displays when the button is clicked



More Example of Hidden Form Field

A Form tag that uses JSP expressions to set hidden field values

```
<form action="cart" method="post">
    <input type="hidden" name="productCode"
       value="<%=product.getCode()%>">
    <input type=text size=2 name="quantity"
       value="<%=lineItem.getQuantity()%>">
    <input type="submit" name="updateButton" value="Update">
</form>
```

Advantages and Disadvantages Hidden Form Field

Advantages of using hidden form field

- It will always work whether cookie is disabled or not.
- Hidden boxes reside in web pages of the browser windows, so they do not provide a burden to the server.

Disadvantages of using hidden form field

- It is maintained at server side.
- Extra form submission is required on each pages.
- Only textual information can be used.
- The hidden box values of the form page can be viewed using the source code of the web page. That means there is no security and are not appropriate for secure data like passwords.

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URL Rewriting

URL Rewriting

- URL rewriting is a method of session tracking in which some extra data (session ID) is appended at the end of each URL.
- This extra data identifies the session. The server can associate this session identifier with the data it has stored about that session.
- This method is used with browsers that do not support cookies or where the user has disabled the cookies.
- In URL rewriting, a token(parameter) is added at the end of the URL as follow:

format : `url?name1=value1&name2=value2`
example : `hello?sessionid=12345&user=ali`

- A name and a value is separated using an equal (=) sign, a parameter name/value pair is separated from another parameter using the ampersand(&).

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Example of URL rewriting

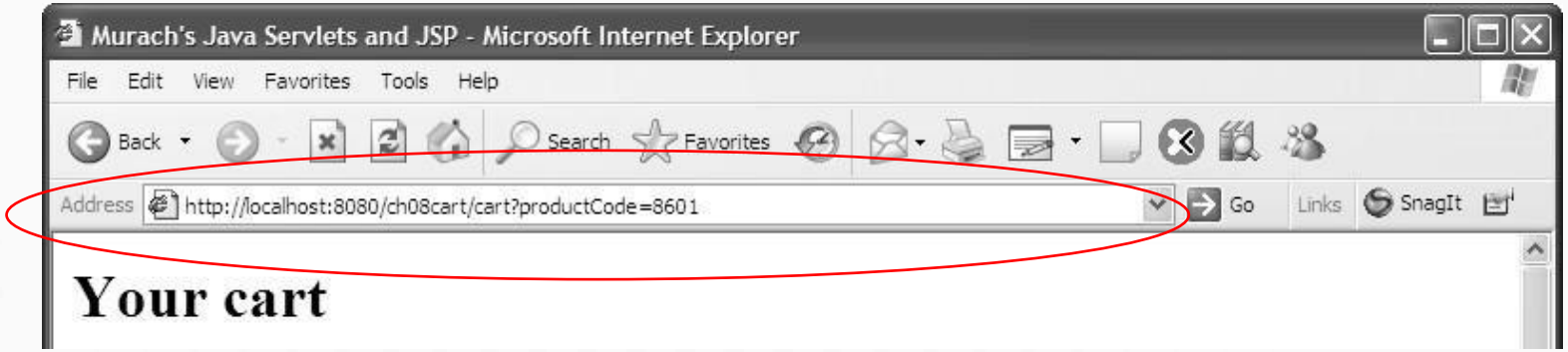
- **An A tag that adds a product code to a URL**

```
<a href="cart?productCode=8601">Add to cart</a>
```

- **The link displayed in a browser**

86 (the band) - True Life Songs and Pictures	$14.95	Add To Cart

- **The URL that displays when you click on the link**



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More examples of URL Rewriting

A Form tag that calls a JSP

```
<form action="cart.jsp?productCode=jr01" method="post">
```

An A tag that uses a JSP expression for the product code

```
<a href="cart?productCode=<%= productCode %>" >
Add to cart</a>
```


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How to use URL encoding to track sessions if cookies is disabled?

- If the user has disabled per-session cookies, you can use ***URL encoding*** to keep track of the ID for the session. To do that, you must convert any relevant HTML pages to JSPs, and you must encode all relevant URLs.

- When you encode a URL, the session ID is passed to the browser in the URL.

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A method to encode a URL

Method	Description
`encodeURL(String url)`	Returns a string for the specified URL. If necessary, this method encodes the session ID in the URL. If not, it returns the URL unchanged.

How to encode a URL in a Form tag

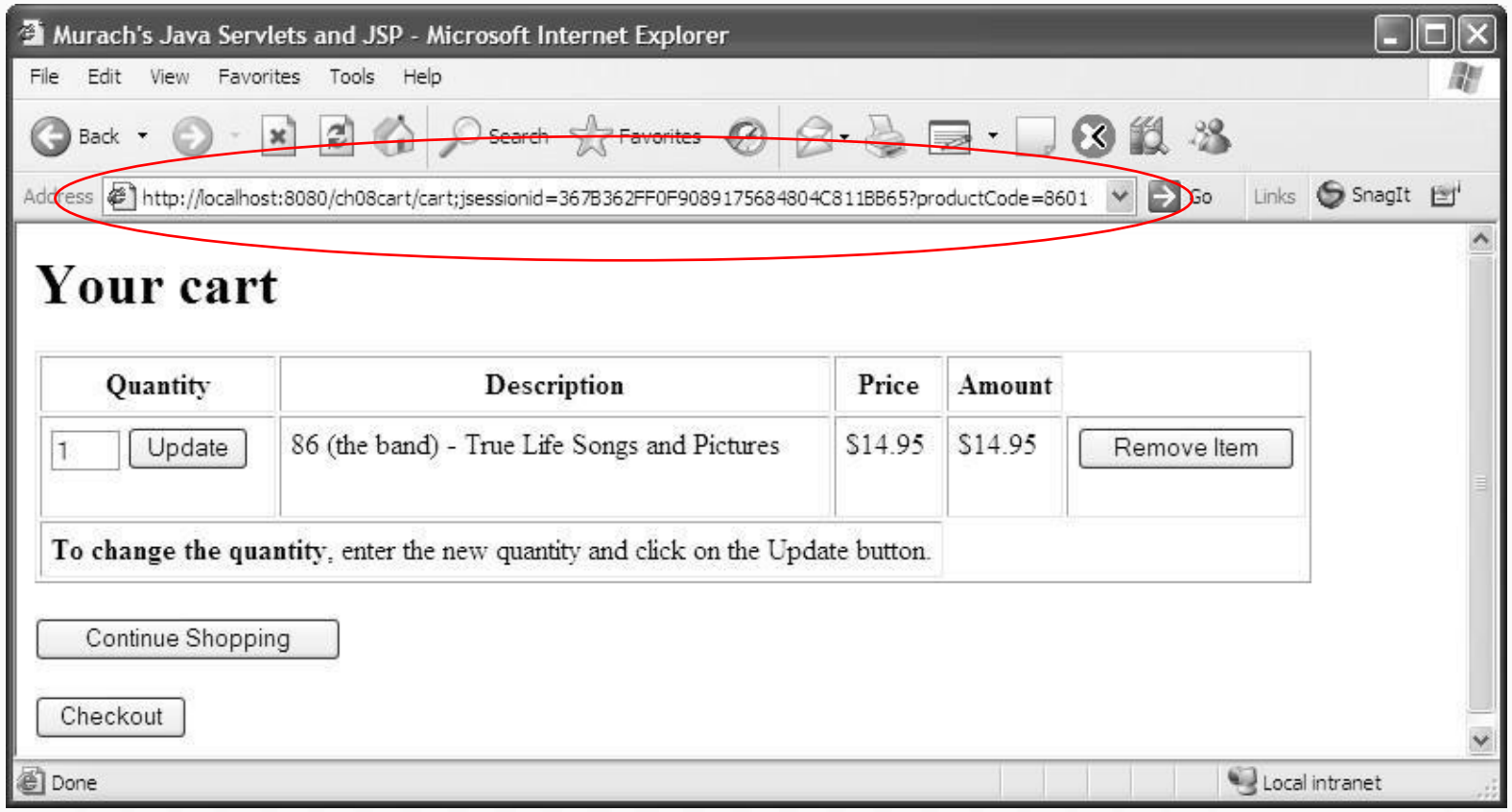
```
<form action="<%=response.encodeURL("cart")%>"
      method="post">
```

How to encode a URL in an A tag

```
<a href="<%=response.encodeURL("cart?productCode=8601")%>">
Add To Cart
</a>
```


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A URL after it has been encoded



Advantages and Disadvantages URL Rewriting

Advantages of using URL Rewriting

- It will always work whether cookie is disabled or not.
- Extra form submission is not required on each pages.

Disadvantages of using URL Rewriting

- It will work only with links.
- It can send only textual information.
- Most browsers limit the number of characters that can be passed by a URL to 2,000 characters.
- It's difficult to include spaces and special characters such as the ? and & characters in parameter values.
- Every time we need to rewrite the URL with session-id value in the generated form, for this we must execute the encoded URL() method.

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HTTP Session - Servlet

- `HttpSession` object is used to store entire session with a specific client. We can store, retrieve and remove attribute from `HttpSession` object.
- Any servlet can have access to `HttpSession` object throughout the `getSession()` method of the `HttpServletRequest` object.
- The `HttpServletRequest` interface provides two methods to get the object of `HttpSession`:
  - **`public HttpSession getSession()`**: Returns the current session associated with this request, or if the request does not have a session, creates one.
  - **`public HttpSession getSession(boolean create)`**: Returns the current `HttpSession` associated with this request or, if there is no current session and create is true, returns a new session.

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HTTP Session - JSP

- By default, JSPs have session tracking enabled and a new `HttpSession` object is instantiated for each new client automatically.
- Disabling session tracking requires explicitly **turning it off by setting the page directive session attribute to false** as follows:

```
<%@ page session = "false" %>
```

- The JSP engine exposes the `HttpSession` object to the JSP programmer through the implicit **session** object.
- Since **session** object is already provided to the JSP programmer, the programmer can immediately begin storing and retrieving data from the object without any initialization or **`getSession()`**.

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Method for request object

Method	Description
`getSession()`	• Returns the HttpSession object associated with this request. If the request is not associated with a session, this method creates a new HttpSession object and returns it.

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Methods for session object

Method	Description
`setAttribute(String name, Object o)`	Stores any object in the session as an attribute and specifies a name for the attribute.
`getAttribute(String name)`	Returns the value of the specified attribute as an Object type. If no attribute exists for the specified name, this method returns a null value.
`removeAttribute(String name)`	Removes the specified attribute from this session.

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How to set and get session attributes?

- A session object is created when a browser makes the first request to a site. It is destroyed when the session ends.
- A session ends when a specified amount of time elapses without another request or when the user exits the browser.
- The session object is a built-in JSP object. As a result, you don’t need to create the session object when working with JSPs.

Examples of code that...

Gets a session object

```
HttpSession session = request.getSession();
```

Sets a String object as an attribute

```
session.setAttribute("productCode", productCode);
```

Sets a user-defined object as an attribute

```
Cart cart = new Cart(productCode);
session.setAttribute("cart", cart);
```

Gets a String object

```
String productCode =
    (String) session.getAttribute("productCode");
```

Gets a user-defined object

```
Cart cart = (Cart) session.getAttribute("cart"); if
(cart == null)
    cart = new Cart();
```

Removes an object

```
session.removeAttribute("productCode");
```


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More methods of the session object

Method	Description
`getAttributeNames()`	Returns a java.util.Enumeration object that contains the names of all attributes in the HttpSession object.
`getId()`	Returns a string for the unique Java session identifier that the servlet engine generates for each session.
`isNew()`	Returns a true value if the client does not yet know about the session or if the client chooses not to join the session.

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More methods of the session object (cont.)

Method	Description
`setMaxInactiveInterval``(int seconds)`	By default, the maximum inactive interval for the session is set to 1800 seconds (30 minutes). To increase or decrease this interval, supply a positive integer value. To create a session that won't end until the user closes the browser, supply a negative integer such as –1.
`invalidate()`	Invalidates the session and unbinds any objects that are bound to it.

Examples of code

A method that gets all the names of the attributes for a session

```
Enumeration names = session.getAttributeNames();
while(names.hasMoreElements())
{
    System.out.println((String) names.nextElement());
}
```

A method that gets the ID for a session

```
String jSessionId = session.getId();
```

A method that sets the inactive interval for a session

```
session.setMaxInactiveInterval(60*60*24); // one day
session.setMaxInactiveInterval(-1);  // until the browser is closed
```

A method that invalidates the session and unbinds any objects

```
session.invalidate();
```


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Examples of code

An example that synchronizes access to the session object

```
Cart cart;
synchronized(session)
{
    cart = (Cart) session.getAttribute("cart");
}
```

Another example that synchronizes access to the session object

```
synchronized(session)
{
    session.setAttribute("cart", cart);
}
```


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A web browser with three windows accessing the same session object



How to provide thread-safe access to the session object?

- Each servlet creates one session object that exists for multiple requests that come from a single client.
- If the client has one browser window open, access to the session object is thread-safe.
- If the client has multiple browser windows open, it's possible (though highly unlikely) that two threads from the same client will access the session object at the same time.
- As a result, the session object isn't completely thread-safe.

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Advantages and Disadvantages HTTP Session

Advantages of using HTTP Session

- There are no restrictions on the size of the object, any kind of object can be stored in a session.
- The usage of the session is not dependent on the client's browser.
- It is secure and transparent.

Disadvantages of using HTTP Session

- Performance overhead in case of large volumes of data/user, because session data is stored in server memory.

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Summary

Summary

- HTTP is a stateless protocol, so web application must provide for session tracking.
- Session tracking enable an application to relate each request to a specific browser and to the data for that session.
- To provide for session tracking, Java creates one session object for each browser. Then, you can add attributes like variables and objects to this session object, and can retrieve the values of these attributes in any of the servlet and JSPs that are run during the session.
- There are four techniques session tracking – cookies, hidden field, URL rewriting and HTTP Session.
- In general, it is considered a best practice to implement session tracking by using cookies. The session ID is stored in a cookie on the user's browser. However, it doesn't work unless the browser enables cookies.
- It's also possible to implement session tracking by using URL encoding. This work even when the browser doesn't enable cookies.
- To pass parameters to a servlet, URL rewriting or hidden fields also can be used.
- HttpSession object is used to store entire session with a specific client.

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TOPIC 5 – Java Web Session Tracking

The End

Credit:
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